



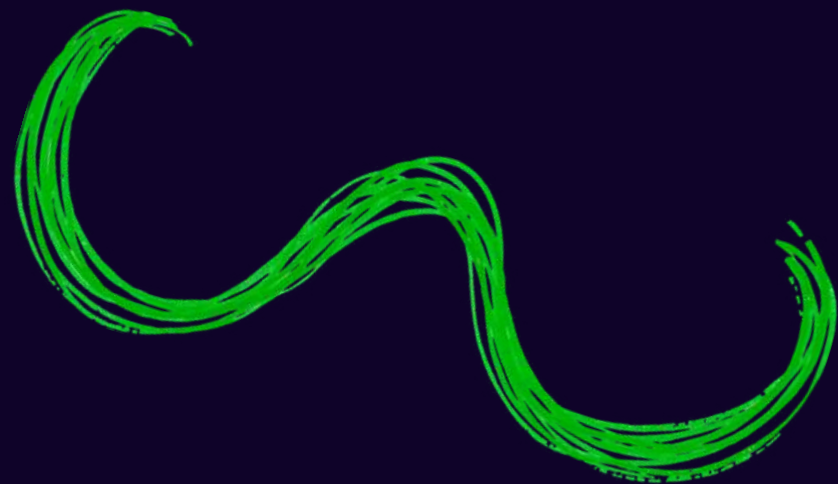
30 MIN



SEED



GERMINATION RACE



LET THE SEED RACE BEGIN!

IS THIS TINY MOONG SEED ALIVE?

DISCOVERING THE MAGIC OF SEEDS

Every morning, Grandmother soaks moong seeds in water. By the next day, tiny white roots appear! These sprouted seeds are called ankurit moong. But wait — how did this happen? Is the seed alive like us?



LOOK! THIS TINY SEED HAS
STARTED TO GROW. IS IT
ALIVE?



See how the moong seed has changed overnight! A tiny white root is peeking out. Something magical is happening inside this little seed.

LET'S FIND OUT TOGETHER HOW SEEDS WAKE UP AND GROW!





WHAT IS GERMINATION?

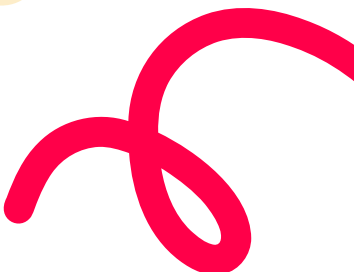
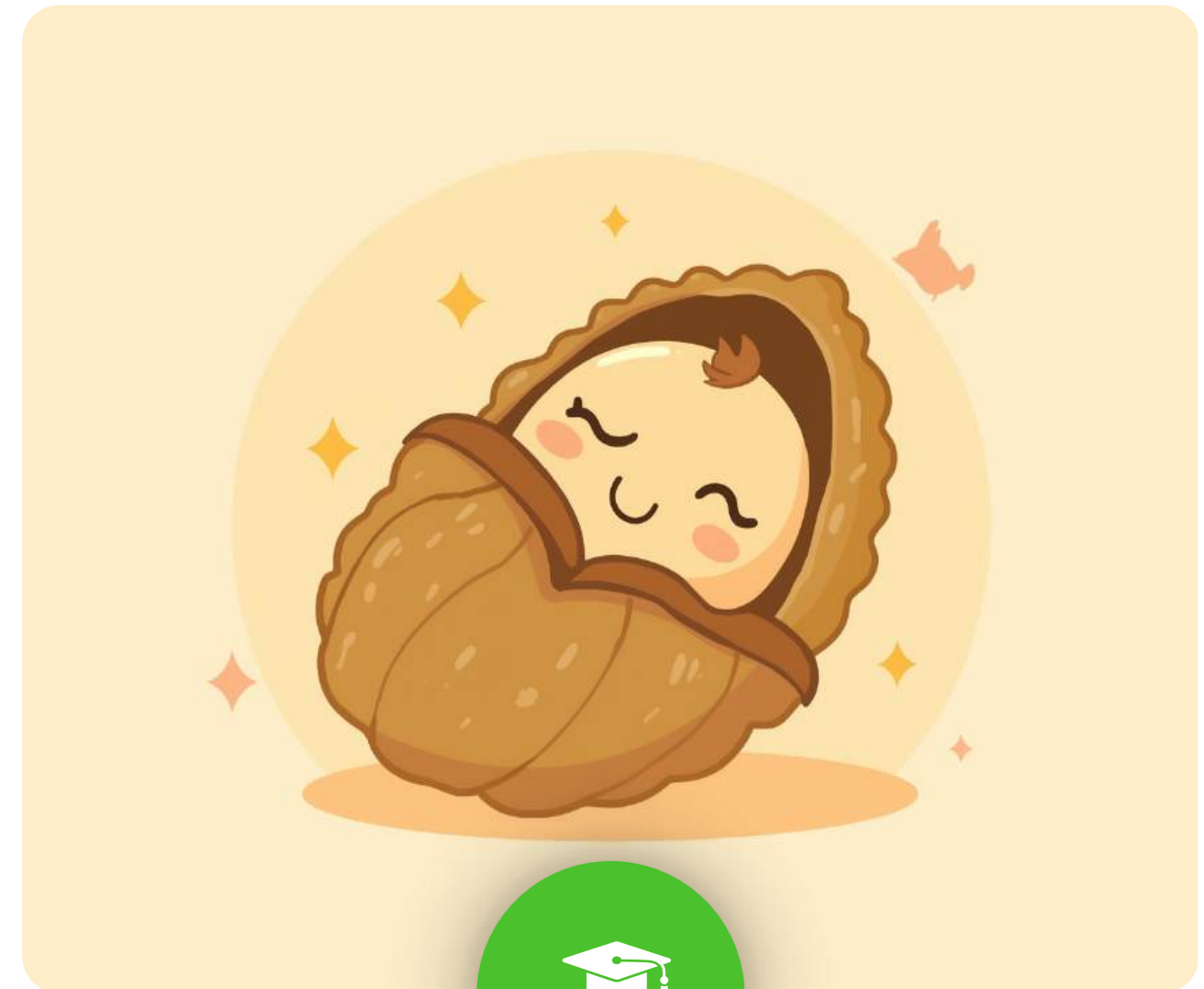
Germination is when a seed wakes up and starts to grow. Think of it like magic happening under the soil! A tiny seed holds everything it needs to become a big, beautiful plant. When conditions are just right, the seed begins its amazing journey of growth.

SEED A SEED IS LIKE A SLEEPING BABY WRAPPED IN A COZY BLANKET

COAT THE SEED COAT PROTECTS THE BABY PLANT INSIDE
UNTIL IT'S READY TO WAKE UP



SEED GERMINATION RACE





SEED GERMINATION RACE

CONDITION 1 – WATER

Seeds need water to wake up. Just like we feel thirsty and need water, seeds need water to start their growing journey!

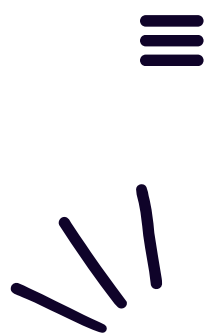


**WATER WAKES UP
SLEEPING SEEDS!**



Water makes the seed swell and come alive. Like Grandmother soaks moong for breakfast sprouts! Seeds drink water overnight and wake up ready to grow.

LIKE GRANDMOTHER SOAKS MOONG FOR BREAKFAST SPROUTS!

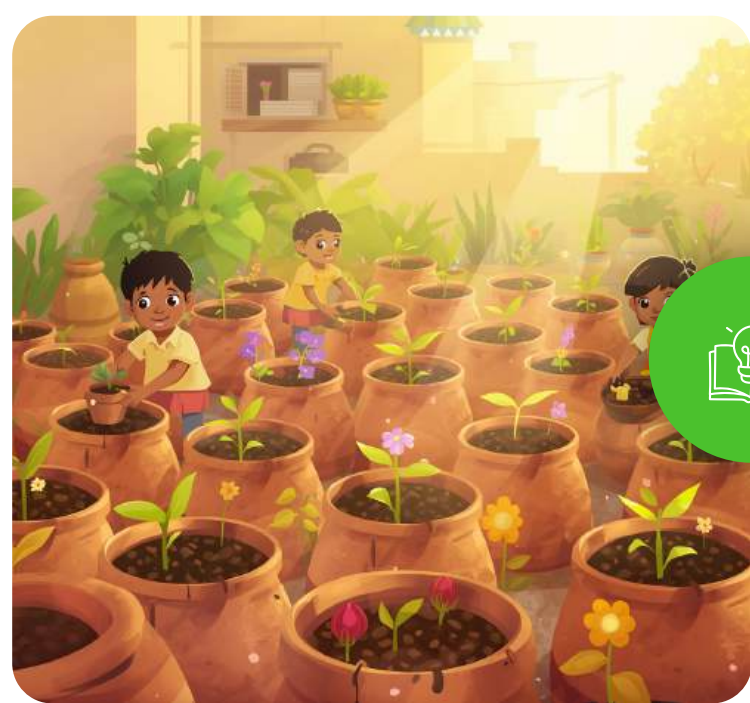


CONDITION 2 – WARMTH

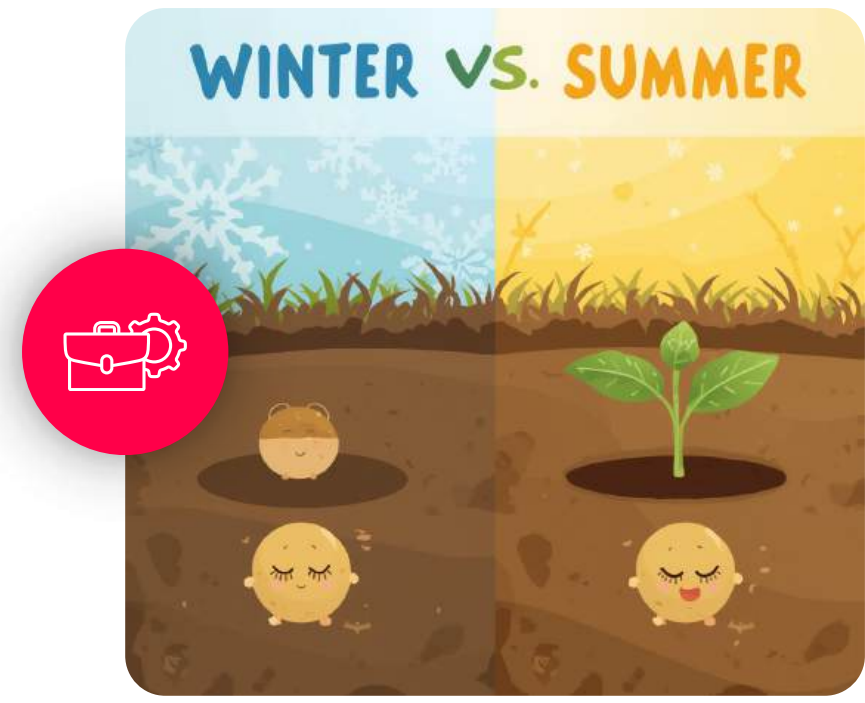
Warmth helps seeds wake up faster. Warm soil is like a cozy bed for the seed. In summer, seeds sprout quickly. In winter, they take longer because the soil is cold.



SEED GERMINATION RACE



WARM SOIL HELPS SEEDS GROW STRONG AND FAST!



Seeds wake faster in warm weather. Just like we feel more active on sunny days, seeds grow better when the soil is warm and cozy!

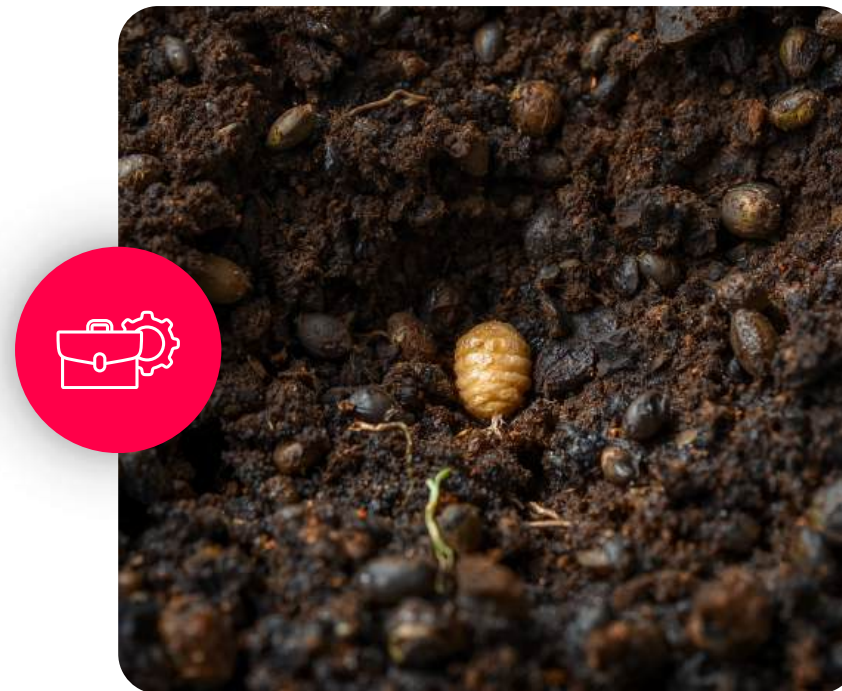
LET'S COMPARE WARM AND COLD SOIL!



SEED GERMINATION RACE

CONDITION 3 – AIR

Seeds need air to breathe and get energy to grow. Just like we need air to breathe, seeds also need oxygen from the air. Without air, seeds cannot wake up and start their growing journey!



**FRESH AIR GIVES SEEDS
THE ENERGY TO SPROUT**

When farmers loosen the soil with their Hand hoe, they create tiny air pockets. These air pockets let seeds breathe underground, giving them the energy they need to push through the soil and grow strong!

AIR HELPS SEEDS BREATHE UNDER THE SOIL!



SEED GERMINATION RACE



STAGE 1

SEED ABSORBS WATER

→ THE SEED WOKE UP WITH WATER!

Just like how we feel thirsty and drink water, seeds also need water to start their journey. The water softens the hard seed coat and gives the tiny plant inside the energy it needs to begin growing. Watch how the moong seed gets bigger!



WATER WAKES IT!



When a seed is placed in water, it absorbs moisture through its seed coat. The seed swells up like a sponge, becoming plump and soft. This is the first step in waking up from its long sleep!

THE SEED DRINKS WATER AND GETS READY TO GROW





SEED GERMINATION RACE

STAGE 2

SEED COAT BREAKS



STAGE 2: SEED COAT BREAKS

The seed coat cracks open like a blanket unwraps. Inside, a tiny white tip peeks out, ready to grow into a new plant!

COAT BREAKS IT!

→ THE BLANKET UNWRAPS!

When the seed drinks enough water, it swells up so much that the outer coat splits open. This is like unwrapping a blanket from a sleeping baby who is ready to wake up and stretch!





SEED GERMINATION RACE

STAGE 3

ROOT COMES OUT



ROOT DIGS DOWN!

Watch closely! The tiny white root pushes downward into the soil. It's searching for water and nutrients to help the plant grow strong and healthy.

THE ROOT DRINKS WATER FROM THE SOIL LIKE TINY FEET!

—————→ **ROOTS ARE THE PLANT'S DRINKING STRAWS!**

The root grows down like the plant's feet, drinking water. Just like how we need our feet to stand strong, plants need roots to hold them in the soil and drink up all the water they need to grow big!





SEED GERMINATION RACE

STAGE

4 – SHOOT PUSHES UP

→ REACHING FOR THE SUNSHINE!

Watch the tiny green shoot push upward from the seed! Just like when you stretch your arms up high in the morning, the shoot reaches for the sky. The shoot will become the stem and leaves of the new plant.



STAGE 4: SHOOT PUSHES UP!



The shoot grows up toward light, like a little hand reaching up. It pushes through the soil to find the warm sunshine above!

SHOOT RISES UP!





SEED GERMINATION RACE



STAGE 5

FIRST LEAVES APPEAR

—————→ **A NEW PLANT BEGINS ITS LIFE!**

When the shoot reaches sunlight, the first green leaves unfold like tiny hands opening up. These leaves are called cotyledons. They help the baby plant make its own food using sunlight. Now the seedling is ready to grow bigger and stronger!

LEAVES OPEN WIDE!

The tiny seedling unfolds its first pair of leaves, ready to catch sunlight and begin making food. This is the start of a new plant's journey!

THE FIRST GREEN LEAVES OPEN WIDE TO START THE PLANT'S LIFE.

WATCH THE MAGIC HAPPEN!





GERMINATION MEMORY RHYME

- Water wakes it, coat breaks it,
- Root digs down, shoot rises up,
- Leaves open wide — a seedling grows!

EASY STAGES TO REMEMBER WITH THIS FUN RHYME!

CATCHY RHYME TO SING ALONG AND NEVER FORGET!



SEED GERMINATION RACE

Seed Germitlation





SEED GERMINATION RACE

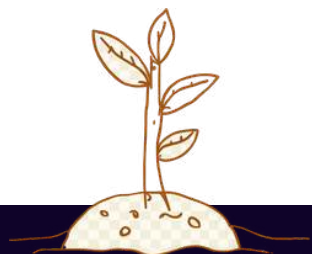
THE GERMINATION RACE



Three seeds enter the race: Moong, Chana, and Wheat
Each seed has a different coat thickness and drinking speed. Who will sprout the fastest? Let's find out!



Just like runners at a starting line, our seeds are ready to race!
Moong is small and quick, Chana is round and steady, and
Wheat takes its time. Watch them grow!



SEED SPEEDS AND WHY



Moong Seeds 1-2 Days

FASTEST! Thin seed coat drinks water quickly. That's why Dadi soaks moong overnight for breakfast sprouts!



Chana Seeds 2-3 Days

Medium speed! Slightly thicker coat takes a bit longer to absorb water and sprout.



(Wheat) 2-4 Days

Slower! Thicker seed coat means it takes more time to wake up and start growing.



REAL-LIFE INDIAN CONNECTIONS



GRANDMOTHER
SOAKING SEEDS



KITCHEN GARDEN SPROUTS



FARMER SOWING SEEDS



SEED GERMINATION RACE



Seeds grow all around us — at home, farms, and markets! From Grandmother's kitchen to the farmer's field, germination is everywhere in our daily lives.



Seeds are magic helpers in our food and festivals! Every sprout you eat and every plant you see started as a tiny seed that woke up with water, warmth, and air.



QUICK RECAP + QUIZ!

TEST WHAT YOU LEARNED ABOUT HOW SEEDS GROW!

Remember our magic chant:

"Water, warmth, and air — three things seeds need
to grow out there!"

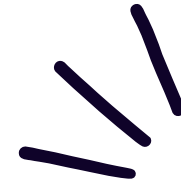
Think carefully about what we learned today
about moong, chana, and Grandmother seeds!



Q1: NAME ONE THING SEEDS NEED TO GROW



Q2: DOES THE ROOT GROW UP OR DOWN?



SEED GERMINATION RACE

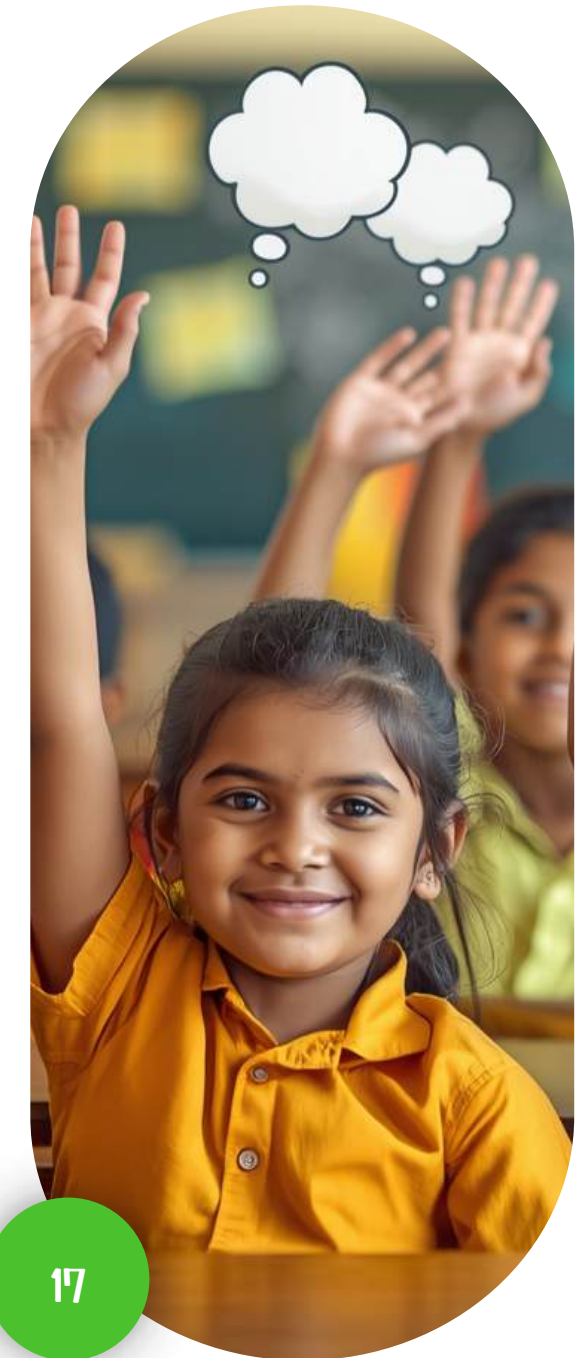


**QUIZ TIME! ANSWER
THESE QUESTIONS**

RAISE YOUR HAND!

16

17



SPROUT MOONG AT HOME IN 3 STEPS!



Step 1: SOAK

Put moong seeds in a bowl, add water, and leave them overnight. The seeds will drink water and get plump!



Step 2: WRAP

Drain the water, wrap the seeds in a wet cloth, and keep them in a warm place. Like a cozy blanket!



Step 3: WATCH

Check daily — see a tiny white root in 1-2 days! Your seeds are growing!



THANK YOU!

Thank you for joining the Seed Germination Race! You learned how seeds wake up with water, warmth, and air. Now you know the magic of how tiny seeds become big plants! Any questions? Let's cheer for our little seeds!

